

Juan David Muñoz-Bolaños

Rechenauerstraße 42, Innsbruck, Austria (Tirol)
+43 676 9115145
juan.munoz@i-med.ac.at

Jenoptik (TRIOPTICS GmbH)

Sophie Krug
Trioptics GmbH
Strandbaddamm 6
22880 Wedel, Germany

Subject: Application for Development Engineer - OMS

Dear Hiring Team,

I am writing to express my strong enthusiasm for the Development Engineer - OMS position at Jenoptik (TRIOPTICS). Concluding my PhD in Optics at the Medical University of Innsbruck and holding an M.Sc. in Photonics, my extensive hands-on experience in precision optical metrology, automated hardware control, and machine vision directly aligns with your requirements for developing next-generation optical measurement systems.

Throughout my academic research and industry tenure at INTECOL S.A.S., I have taken sole responsibility for designing complex opto-mechatronic systems from the ground up. I routinely integrate custom optics, spatial light modulators, and sensitive detection modules, which mirrors the precision required for assembling and calibrating modules like your ATS alignment turning stations and autocollimators. My background in industrial image processing further equips me to develop new machine vision algorithms for optical parameter measurement.

Python Automation & Hardware Synchronization:

Modern optical metrology relies on a seamless bridge between precision hardware and analytical software. I develop high-performance Python pipelines to synchronize data acquisition across spatial modulators, detection units, and various motion stages. My ability to design and program automated calibration processes ensures that complex physical measurements are both repeatable and reproducible.

Integration & Availability:

Having lived and worked as an active researcher in Austria and Germany for the past six years, I hold a certified B2 level in German alongside fluent English. I am fully integrated into the DACH work culture and am prepared to immediately transition my residency for this full-time role in Wedel.

With my deep passion for precision engineering, advanced Python programming, and a strong foundation in optical metrology, I am highly motivated to drive innovation at Jenoptik. I welcome the opportunity to discuss how my hands-on prototyping and machine vision skills can directly advance your ATS product line.

Sincerely,

Juan David Muñoz-Bolaños

Optical System Engineer
Medical University of Innsbruck

Juan David Muñoz-Bolaños

Optical System Engineer

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PROFESSIONAL SUMMARY

Experimental Physicist and Optical Engineer specialized in the architecture, integration, and calibration of advanced opto-mechatronic platforms and measurement systems. Extensive hands-on laboratory experience integrating high-power lasers, precision metrology, spatial modulators, and highly sensitive detection modules. Proven capability to bridge physics with robust hardware control software (Python) to drive automated test processes, machine vision algorithms, and conceptual research to reliable industrial prototypes.

TECHNICAL COMPETENCIES

Optical Metrology & Machine Vision

Substantial expertise in experimental optics, custom system design, and industrial image processing. Demonstrated ability to assemble, calibrate, and program complex automated test setups, directly analog to autocollimators and alignment turning stations.

Prototyping & Hardware Integration

Hands-on integration of continuous-wave/pulsed lasers, spatial light modulators, motorized motion stages (rotation/linear), and highly sensitive detection modules. Focus on optimizing signal-to-noise ratios and measurement repeatability.

Software Integration & Control

Advanced Python for hardware interfacing, data acquisition, and automated test sequence programming. Proficient in integrating Traditional AI and machine vision to synchronize opto-mechatronic hardware with precise measurement protocols.

PROFESSIONAL EXPERIENCE

Medical University of Innsbruck

Innsbruck, Austria

PhD Researcher - Optical Engineering

Jan 2022 – Present

- **System Architecture & Alignment:** Designed and constructed advanced Two-Photon microscopy setups from the ground up, directing the full optical integration from laser source to highly sensitive detectors.
- **Automated Test Sequences:** Developed rigorous data acquisition and automated calibration pipelines using Python, coordinating complex measurement protocols across motion stages and electro-optical components.
- **Core Technology Integration:** Integrated and characterized novel core technologies, including adaptive optics hardware, applying vectors and matrix mathematics to fundamentally improve spatial resolution.
- **Project Management:** Led independent R&D initiatives, taking sole operational responsibility for evaluating new concepts and maintaining complex laboratory prototypes.

Leibniz Institute of Photonic Technology (IPHT)

Jena, Germany

R&D Assistant Researcher

Jun 2020 – Dec 2021

- **Prototype Construction:** Designed and assembled a dispersive 1064 nm fiber probe Raman imaging spectrometer, and calibrated vibrational point-scanning prototypes for clinical tissue applications.
- **Innovation & Evaluation:** Bridged theoretical R&D with practical lab implementation, utilizing traditional AI (supervised tools) for hyperspectral data analysis to streamline experimental characterization.

INTECOL S.A.S.

Medellin, Colombia

Machine Vision System Engineer

Mar 2018 – Mar 2019

- **System Architecture & Component Selection:** Directed the physical design of optical inspection setups by analyzing and selecting appropriate sensors, industrial cameras, illumination schemes, and custom optics.
- **Machine Vision Integration:** Implemented robust, industrial-grade image processing pipelines to ensure precise real-time defect detection and tracking reliability within manufacturing environments.

EDUCATION

Medical University of Innsbruck
Ph.D. in Adaptive Optics & Microscopy

Austria
Jan 2022 – Present

Friedrich Schiller University Jena
M.Sc. in Photonics

Germany
2019 – 2021

University of Cauca
B.Sc. in Physics Engineering

Colombia
2012 – 2018

PUBLICATIONS & AWARDS

- **Awards & Selections:** Selected for ASML and ZEISS Summer Schools; awarded COLFUTURO Grant 2026 for the best Colombian professionals.
- Muñoz-Bolaños, J. D., et al. Design of a dispersive 1064 nm fiber probe Raman imaging spectrometer and its application to human bladder resectates. *Applied Sciences*, 14(11), 4726 (2024).
- Muñoz-Bolaños, J. D. et al. Confocal Raman Microscopy with Adaptive Optics. *ACS Photonics* 12(1), 176–184 (2025).
- Sohmen, M., Muñoz-Bolaños, J. D., et al. Optofluidic adaptive optics in multi-photon microscopy. *Biomedical Optics Express* 14(4), 1562–1578 (2023).

LANGUAGES

English (Proficient),
German (Intermediate/Working Proficiency),
Spanish (Native)

REFERENCES

Prof. Monika Ritsch-Marte
Head of Institute for Biomedical Physics
monika.ritsch-marte@i-med.ac.at

Medical University of Innsbruck

Dr. Christoph Krafft
Group leader: Spectroscopy Imaging
christoph.krafft@leibniz-ipht.de

Leibniz Institute of Photonic Technology

Passports № / Passport No

MUNOZ BOLANOS

COLOMBIANA

Sexo / Sex
Lugar de nacimiento / Place of birth

M LA CRUZ COL

Fecha de expedición / Date of issue

11 DIC/DEC 2018

Fecha de vencimiento / Date of expiry

10 DIC/DEC 2028

Autocidad / Authority

G. ANTIOQUIA

Firma del titolare / Titolare's signature

Dear Family

[illegible]



URKUNDE

Die Friedrich-Schiller-Universität Jena verleiht durch die
Physikalisch-Astronomische Fakultät
mit dieser Urkunde

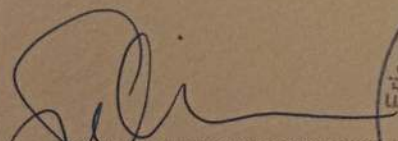
Herrn Juan David Munoz Bolanos
geb. 04.06.1994 in La Cruz (Kolumbien)
den Hochschulgrad

MASTER OF SCIENCE (M.Sc.)

Er hat die Masterprüfung (120 ECTS)
im Studiengang »Photonics«

bestanden.

Jena, 20.12.2021


Prof. Dr. Christian Spielmann
Dekan




Prof. Dr. Silvana Botti
Vorsitzende des
Prüfungsausschusses

Zertifikat

Muñoz-Bolaños JUAN DAVID

geboren am 04.06.1994

in La Cruz / Kolumbien

hat das Modul **Schriftliche Prüfung**

ÖSD Zertifikat B2

am Prüfungszentrum

BFI Tirol Bildungs GmbH

in Innsbruck / Österreich

bestanden



Datum: 16.09.2025

für das ÖSD: Prüfungsvorsitzende/r
Maria VALLAZZA



ÖSD ist Mitglied von



ö s d

österreich schweiz deutschland

ID-Nummer: ZB2Ö25s34291

Juan Munoz Bolanos

has participated in the

**ZEISS European
Summer School**

Lithography Optics &
Semiconductor Technologies

26 - 27 June 2025

ZEISS Semiconductor Manufacturing Technology, Oberkochen, Germany

Thomas Marschner

Dr. Thomas Marschner
Program Manager Funding Projects

Thomas Stämmler

Dr. Thomas Stämmler
CTO & Head of Product Strategy



AGREEMENT FOR GUESTS

to carry out a practical training

with

Mr. **Juan David Munoz Bolanos** Date of birth: **1994-06-04**
temporarily living at **Emil-Wölk-Str. 7, 07747 Jena** (guest),

Student of the Friedrich-Schiller-University Jena, Abbe School of Photonics, gets in execution of his practical course in the research department Spectroscopy and Imaging, working group Raman and Infrared Histopathology (0104) from 2021-01-01 up to 2021-09-30 admission to the necessary rooms within Leibniz-IPHT.

The use of further devices, media and other infrastructure takes place in agreement with the Heads of the respective departments.

Mr. Juan David Munoz Bolanos has to observe the security regulations for working and other regulations valid in Institute as well as instructions of the laboratory and clean room responsible persons of the Leibniz-IPHT.

Mr. Juan David Munoz Bolanos has to treat all affairs and procedures strictly confidential. Copies, duplicates and others of business and technical information of each kind are forbidden, unless these are sanctioned by the responsible Department Heads.

Publications, reports and lectures about topics which belong together with the fields of work of Leibniz-IPHT require the previous written agreement of the Executive Board.

The liability of Mr. Juan David Munoz Bolanos is limited to cases of wilful action and gross negligence.

Mr. Juan David Munoz Bolanos is insured against accidents on the Friedrich-Schiller-University Jena, Abbe School of Photonics in the context of his/her practical course during his/her length of stay at Leibniz-IPHT.

Jena, 13 November 2020


Prof. J. Popp
Chairman
of Leibniz-IPHT


F. Sondermann
Executive Member
of Leibniz-IPHT


Juan David Munoz Bolanos
Guest

INTECOL

OPTIMIZING THE RHYTHM OF THE WORLD

Medellin, February 4, 2019

TO WHOM IT MAY CONCERN

Integración Tecnológica Industrial Intecol S.A.S. with Tax ID (NIT) 811.041.410-4 certifies that Mr. JUAN DAVID MUÑOZ BOLAÑOS, identified with Citizenship ID No. 1.061.772.278, has been employed by this company since July 3, 2018, under an indefinite term contract, serving in the position of Junior Project Engineer;

earning a salary of One million five hundred thousand pesos (\$1,500,000) plus three hundred thousand pesos (\$300,000) mobility allowance.

I will gladly address any additional information at the phone number 316 3322.

We appreciate your attention,

INTECOL

NIT. 811.041.410-4 - Tel.: 3163822

LENY ADRIANA RUIZ PINO

Administrative and Financial Director

Integración Tecnológica Industrial Intecol S.A.S.
www.intecol.com.co
Medellin Calle 42 #63c-82
PBX. (574) 316 3322 Cell: (57) 301 430 2532